WORKED SOLUTION

STEP I 1992, Question 12

Mechanics • Friction

STEP I 1992, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q12

The key idea

The two ladders are light, so the only way to get a useful equation for one of them is to take moments about the hinge A: that removes the unknown force from the other ladder and from the load at A. The feet want to slide apart, and the outward push is resisted partly by friction and partly by the rod. At the moment of collapse friction is doing all it can, so the rod has to supply the rest.

Solution

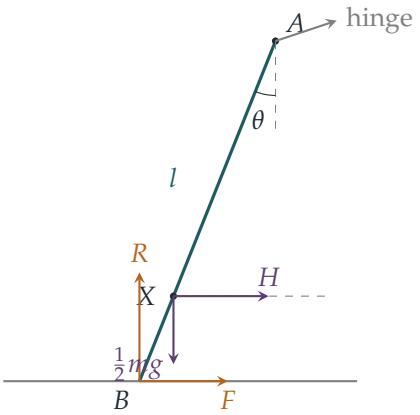
The model

The ladders are light rods, smoothly hinged at A. The wooden rod XY is uniform, of mass m, and pin-jointed at X and Y. The mass M is a particle at A. Write $t = \tan\theta$. By the left–right symmetry of the whole arrangement, the two feet carry equal normal reactions R, and resolving vertically for the whole step-ladder gives

$$2R = (M+m)g \quad \Longrightarrow \quad R = \tfrac{1}{2}(M+m)g.$$

Since the ladders lean outwards, the feet tend to slide apart, so the friction F at each foot acts inwards.

Forces from the rod. The rod is horizontal and has its own weight, so the forces at its ends are *not* along it. By symmetry each end carries half its weight, so the ladder exerts an upward force $\frac{1}{2}mg$ on the rod at X, together with a horizontal force H outwards (the tension). By Newton's third law, the rod pulls ladder AB at X with a horizontal force H *inwards* and a vertical force $\frac{1}{2}mg$ *downwards*.



Moments about A for ladder AB

Measured from A, the foot B is $l\sin\theta$ across and $l\cos\theta$ down, and X is $\frac{3}{4}l\sin\theta$ across and $\frac{3}{4}l\cos\theta$ down. Taking moments about A (all forces at A have no moment):

- R turns the ladder one way with moment $R\,l\sin\theta$;
- F and H (both inwards, acting below A) turn it the other way, with moments $F\,l\cos\theta$ and $H\cdot\frac{3}{4}l\cos\theta$;
- the downward $\frac{1}{2}mg$ at X acts on the same side as R but in the opposite sense, with moment $\frac{1}{2}mg\cdot\frac{3}{4}l\sin\theta$ opposing R.

So

$$F\,l\cos\theta+\tfrac{3}{4}H\,l\cos\theta=R\,l\sin\theta-\tfrac{3}{8}mg\,l\sin\theta,$$

and dividing by $l\cos\theta$ and substituting for R,

$$F+\tfrac{3}{4}H=t\left(\tfrac{1}{2}(M+m)g-\tfrac{3}{8}mg\right)=t\left(\tfrac{1}{2}M+\tfrac{1}{8}m\right)g. \qquad (*)$$

Ask yourself Friction and the rod both hold the feet in. Which value of F do we use?

Equation $(*)$ says friction and tension share a fixed amount of work: the more friction provides, the less the rod needs to. So the step-ladder survives as long as some allowed friction ($F\le\mu R$) leaves $H\le T$, and the smallest tension possible comes from the largest friction, $F=\mu R$. This is the limiting state at collapse that the question lets us assume. Putting $F=\mu R=\frac{1}{2}\mu(M+m)g$ into $(*)$:

$$\begin{aligned}\tfrac{3}{4}H&=t\left(\tfrac{1}{2}M+\tfrac{1}{8}m\right)g-\tfrac{1}{2}\mu(M+m)g\\&=\tfrac{1}{2}(t-\mu)Mg+\tfrac{1}{8}(t-4\mu)mg,\end{aligned}$$

so

$$H=\tfrac{2}{3}(t-\mu)Mg+\tfrac{1}{6}(t-4\mu)mg.$$

Since $t > \mu$, H increases with M: without the rod, friction alone could not hold even the load (with $m = 0$ we would need $t \le \mu$).

The largest safe mass

The rod survives provided $H \le T$, that is

$$\tfrac{2}{3}(t-\mu)Mg \le T - \tfrac{1}{6}(t-4\mu)mg,$$

and dividing by the positive quantity $\tfrac{2}{3}(t-\mu)g$,

$$\boxed{M \le \frac{6T - (\tan\theta - 4\mu)mg}{4(\tan\theta - \mu)g}}$$

So the greatest mass that can safely be placed at A is $\dfrac{6T - (\tan\theta - 4\mu)mg}{4(\tan\theta - \mu)g}$.

Failure under its own weight

With nothing at A ($M = 0$), the tension needed is $H = \tfrac{1}{6}(t-4\mu)mg$, so the rod breaks if

$$\tfrac{1}{6}(\tan\theta - 4\mu)mg > T \iff \tan\theta > \frac{6T}{mg} + 4\mu,$$

as required. In that case the bound on M above is negative: no load at all is safe, because the step-ladder cannot even support itself.

Checks

- If T is huge the safe mass is huge, and if $\tan\theta$ only just exceeds μ the denominator is tiny and again almost any load is safe, because friction nearly suffices on its own.
- A numerical solution of the full statics (both ladders, the rod and the pin at A, with twelve unknown force components and limiting friction at both feet) for 2000 random configurations reproduced the tension formula to about 10^{-12}. For example, with $\theta = 45^\circ$, $\mu = 0.2$, $m = 2\,\mathrm{kg}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$, the rod tension with $M = 0$ is $\tfrac{1}{6}(0.2)(19.6) \approx 0.653\,\mathrm{N}$.

Common slips

- Treating the rod as a light tie, so the force at X is purely along XY. The rod has weight, so each end also carries $\tfrac{1}{2}mg$ vertically, and this is where the $\tfrac{1}{8}m$ and the 4μ come from.
- Putting friction outwards at the feet. Since $\tan\theta > \mu$, the feet tend to splay, so friction points inwards and helps the rod.

Beyond STEP

A frame like this, where the equations of equilibrium alone don't fix every force (here, friction and tension can trade off), is called *statically indeterminate*. Engineers resolve such cases with extra information: a limiting condition, as here, or the elasticity of the members. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen your understanding of forces, torques and equilibrium.

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